Policy-Private Consent Enforcement on Public Blockchains via TPS-Assisted Zero-Knowledge Verification

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***Abstract*—In blockchain-based healthcare data sharing, consent policies are expected to be enforced in a verifiable and auditable manner. However, public blockchain transparency may expose sensitive policy information through on-chain representations, verification interfaces, and observable execution artifacts. Existing privacy-preserving designs mainly protect data contents, user identities, or transaction details, but they insufficiently address the confidentiality of governing consent policies and policy-dependent leakage during enforcement.**

This paper studies policy-private consent enforcement for blockchain healthcare systems. We propose the Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter (TPS-ZKPI), which enables authorization decisions to be verified while limiting leakage about policy content, policy structure, requester attributes, and public-chain metadata. In TPS-ZKPI, the provider commits to a hidden consent policy on-chain and provisions policy-bound proving material to a Trusted Proof Service (TPS). The TPS constructs the witness from requester-submitted credentials and access context, generates a zero-knowledge proof of policy compliance, and the proof is verified on-chain against the policy commitment.

TPS-ZKPI adopts an explicit off-chain trust boundary for confidential policy handling. The TPS has access to policy-bound proving material and requester authorization evidence, while zero-knowledge verification provides public auditability of authorization correctness without exposing the private witness to the blockchain. To reduce policy-dependent leakage through public verification artifacts, TPS-ZKPI combines fixed-length policy encoding, a reusable zero-knowledge policy interpreter, commitment-based policy binding, and public-chain transcript normalization.

We evaluate TPS-ZKPI against metadata-only adversaries that observe public-chain artifacts and conduct inference attacks on policy content, policy distinguishability, and policy structure and parameters. Under a multi-seed permutation-test criterion, no modeled supervised or unsupervised attack recovers hidden policy information above the chance baseline, and a paired gas-padding ablation attributes this protection specifically to transcript normalization at a 38–40% gas overhead.

***Index Terms*—Blockchain, policy privacy, data use ontology, healthcare data sharing**

I. Introduction

BLOCKCHAIN technology has emerged as a promising foundation for decentralized and auditable healthcare data sharing [1]–[3]. Its transparency and immutability enable verifiable consent enforcement and traceable authorization decisions, supporting cross-institutional accountability. At the same time, consent enforcement on public blockchains may expose policy-dependent contextual information, even when the underlying health data remain encrypted [4], [5].

Consent policies in healthcare are often privacy-sensitive [6]. They may encode research purposes, disease scopes, requester eligibility, institutional restrictions, ethics approval requirements, or other contextual conditions. For example, a policy based on biomedical data-use terms may reveal whether a dataset is restricted to disease-specific research, whether ethics approval is required, or whether access is limited to particular institutions or projects. If such policies are directly represented in smart contracts, embedded in verification logic, or reflected in public execution traces, unauthorized observers may infer information about the protected resource, the data owner's constraints, or the requester's authorization context.

Existing privacy-preserving blockchain systems mainly protect data contents, user identities, or transaction details [7]–[9]. Encrypted data-sharing mechanisms such as attribute-based encryption and proxy re-encryption protect access to data [10], [11], while zero-knowledge transaction systems hide asset ownership, values, or participant relationships [12]. These mechanisms are important, but they do not directly address the confidentiality of the policies that govern access decisions. In blockchain-based consent management, policy leakage may arise not only from explicit policy disclosure, but also from policy-dependent artifacts such as calldata size, public-input layout, emitted events, verifier interfaces, and gas usage. Therefore, policy-private consent enforcement requires more than hiding data or proving correctness in zero knowledge; it also requires controlling what the public-chain transcript reveals about the hidden policy.

This paper studies policy-private consent enforcement for blockchain-based healthcare data sharing under public-chain observability. The core problem is how to make consent enforcement publicly verifiable while preventing metadata-only public-chain observers from inferring the governing policy or requester authorization evidence from policy-dependent execution artifacts. In particular, an on-chain verifier should be able to check that a requester satisfies a provider-defined consent policy, while a metadata-only public-chain adversary learns minimal information from the resulting public verification transcript. We focus on Boolean-based consent policies, including policies derived from biomedical data-use conditions such as Data Use Ontology (DUO) terms [6] and disease-scope constraints.

To address this problem, we propose the Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter (TPS-ZKPI). In TPS-ZKPI, the data provider commits to a hidden consent policy on-chain and provisions the corresponding

policy-bound proving material to a Trusted Proof Service (TPS). The requester submits policy-relevant authorization evidence, such as attributes, credentials, and access context, to the TPS. The TPS constructs the private witness and generates a zero-knowledge proof showing that the requester satisfies the committed policy. The proof is then verified on-chain against the provider's policy commitment and fixed-format public inputs, allowing the system to record an auditable authorization outcome without exposing the policy logic in public smart-contract execution.

TPS-ZKPI combines the following design elements. First, it uses TPS-assisted proof generation to keep confidentiality-sensitive policy material and requester evidence outside the public blockchain transcript. Second, it compiles heterogeneous Boolean-based consent policies into fixed-length Reverse Polish Notation (RPN) programs. Third, it evaluates these programs through a reusable zero-knowledge policy interpreter, avoiding policy-specific verifier contracts and supporting policy evolution without changing the public verification interface. Fourth, it applies observability-aware transcript normalization to reduce policy-dependent variation in calldata, logs, public parameters, and gas-related behavior.

We evaluate TPS-ZKPI using attack-driven empirical analysis under a metadata-only adversarial model. The evaluation operationalizes policy privacy through three objectives: Policy Content Hiding (PCH), Policy Indistinguishability (PI), and Structure and Parameter Privacy (SPP). Supervised and unsupervised attacks are used to test whether observable blockchain artifacts allow adversaries to recover hidden policy semantics, distinguish policy families, or infer structural properties. The results show that, after transcript normalization, no modeled attack recovers hidden policy information above the chance baseline under a multi-seed permutation criterion, and a paired ablation isolates gas-transcript normalization as the mechanism responsible.

This paper makes the following contributions:

- We formulate policy-private consent enforcement as a public-chain observability problem in blockchain-based healthcare data sharing, focusing on leakage of policy content, policy distinguishability, and policy structure from public verification artifacts.
- We propose TPS-ZKPI, a TPS-assisted zero-knowledge approach that supports publicly verifiable and auditable consent enforcement while keeping policy-bound proving material and requester authorization evidence off-chain.
- We design a reusable zero-knowledge policy interpreter for fixed-length Boolean policy programs, combining policy commitment binding with a stable on-chain verification interface.
- We provide an attack-driven empirical evaluation of residual policy-dependent leakage from normalized public-chain metadata, covering supervised classification and unsupervised clustering attacks under a multi-seed permutation-test criterion.
- We conduct a paired gas-padding ablation on a public testnet that isolates transcript normalization as the mechanism responsible for the observed protection and quantifies its gas-cost overhead.

The remainder of this paper is organized as follows. Section II discusses related work and positions TPS-ZKPI against existing privacy-preserving blockchain, hidden-policy, reusable proving, and metadata-leakage approaches. Section III defines the system model, trust assumptions, adversarial model, and privacy objectives. Section IV presents the TPS-ZKPI design. Section V reports the empirical evaluation. Section VI discusses limitations and future work, and Section VII concludes the paper.

II. Related Work

Existing work addresses important fragments of policy-private consent enforcement, but not their combination under public-chain observability. Encrypted data-sharing mechanisms, such as Ciphertext-Policy Attribute-Based Encryption (CP-ABE) and proxy re-encryption (PRE) [13], [14], protect access to data. However, CP-ABE-style ciphertext policies may expose policy-related structures, while PRE-based delegation may reveal authorization patterns outside the protection target of data encryption. Transaction-privacy systems, including Zerocash-style protocols [15] and Ethereum-oriented systems such as ZETH and AZTEC [7], [8], provide public verification over private transaction witnesses, but they mainly target asset ownership, values, or participant relationships rather than compliance with a data-owner-controlled consent policy.

Hidden-policy access-control work reduces direct policy exposure by using encrypted policies, off-chain matching, or policy-holder components [16], [17]. However, such designs may still leak policy-shaped artifacts through ciphertext structures, policy-update transactions, authorization workflows, repeated interactions, or privileged policy-holder behavior. Moreover, many such systems are designed for permissioned or off-chain settings, where the observability assumptions are weaker than those of public blockchains.

Privacy-preserving access-control and policy-evaluation mechanisms have been studied in credential-based and XACML/SAML settings [18], as well as in multi-party policy evaluation using homomorphic encryption and secure function evaluation [19]. These approaches support confidential authorization computation, but they are not primarily designed for settings where policy-evaluation artifacts are permanently and globally visible through a public blockchain. In public-blockchain consent enforcement, calldata, public inputs, emitted events, timing, and gas usage are observable to any party. In addition, these approaches do not directly provide compact evidence that a smart contract can verify against a committed, data-owner-controlled policy instance.

Reusable proving systems, including universal-setup SNARKs, composable proof systems, and zkVMs [20]–[22], improve maintainability by reducing the need for policy-specific circuits or verifier contracts. However, when deployed for public-chain consent enforcement, reusable proving alone does not guarantee policy privacy: policy-dependent program representations, public inputs, calldata layouts, logs, or gas profiles may still remain observable. This motivates TPS-ZKPI's use of a domain-specific reusable

TABLE I
POSITIONING OF REPRESENTATIVE APPROACHES AGAINST POLICY-PRIVATE CONSENT ENFORCEMENT UNDER PUBLIC-CHAIN OBSERVABILITY. "YES" INDICATES EXPLICIT SUPPORT, "PARTIAL" INDICATES INDIRECT OR INCOMPLETE SUPPORT, AND "NO" INDICATES THAT THE DIMENSION IS NOT ADDRESSED.

Approach	Policy confidentiality	Metadata-leakage mitigation	Verifiable auditability	Main limitation
Data / transaction privacy [7], [8], [13]–[15]	Partial	Partial	Partial	Protects encrypted data access or transaction semantics, but not the governing consent policy or policy-dependent authorization workflow.
Hidden-policy access control [17]	Partial	Partial	Partial	Reduces direct policy exposure, but may leak through workflows, updates, or policy-holder interactions.
MPC / HE policy evaluation [18], [19]	Yes	No	Partial	Supports confidential computation, but usually lacks public-chain-verifiable authorization evidence.
Universal SNARKs / zkVMs [20]–[22]	No	Partial	Yes	Supports proof reuse and public verification, but does not by itself hide policy-dependent public artifacts.
Metadata-leakage defenses [23]	No	Yes	No	Models or mitigates observable signals, but does not provide hidden-policy consent enforcement.
TPS-ZKPI	Yes	Yes	Yes	Provides policy-private, metadata-aware, verifiable consent enforcement under the stated TPS trust assumption.

policy interpreter, which preserves proof reuse while exposing only a constrained public verification interface.

Public-chain metadata-leakage studies show that cryptographic privacy guarantees may be weakened by observable transaction-level artifacts such as timing and volume patterns [23]. Techniques such as batching, decoys, and traffic shaping can reduce some observable signals, but they are usually designed for payment privacy, anonymity sets, or transaction-linkability settings. They do not directly provide hidden-policy compliance verification or auditable consent enforcement. TPS-ZKPI incorporates this insight by treating metadata leakage as part of policy privacy and by normalizing calldata, logs, public parameters, and gas-related behavior.

Table I summarizes the positioning of TPS-ZKPI. Existing approaches protect data, transactions, authorization computation, proof reuse, or metadata in isolation. In contrast, TPS-ZKPI combines policy confidentiality, metadata-leakage control, verifiable auditability, and a stable verification interface for evolving Boolean-based consent policies.

III. PROBLEM AND THREAT MODEL

A. System and Trust Model

We consider a blockchain-based healthcare data-sharing system in which access to protected resources is governed by provider-defined consent policies. The system involves five roles: a data provider or data owner, a data requester, a Trusted Proof Service (TPS), an on-chain access-control manager (ACM), and a resource access layer. The provider defines a consent policy P for a protected resource and registers an on-chain commitment h to the encoded policy. The requester seeks access by submitting policy-relevant authorization evidence, such as attributes, credentials, and access context. The TPS constructs the private witness and generates a zero-knowledge proof of policy compliance. The ACM verifies the proof on-chain and records the authorization outcome, while the resource access layer grants access according to the recorded decision.

TPS-ZKPI relies on the following assumptions. First, policies are provider-controlled: the provider is responsible for defining the applicable consent policy and committing the corresponding policy instance on-chain. Second, the TPS is trusted to preserve the confidentiality of policy-bound proving material and requester-submitted authorization evidence, and to follow the prescribed proof-generation workflow. Hidden policies and requester evidence are therefore protected from the public blockchain transcript, but they are visible to the TPS. Third, the commitment scheme is assumed to be hiding and binding, and the zero-knowledge proof system is assumed to satisfy completeness, soundness, and zero knowledge with respect to the implemented policy-compliance relation. Credential issuance, revocation, and real-world identity verification are treated as external functions.

B. Public-Chain Adversary

The primary adversary is a passive public-chain observer. The adversary can inspect smart-contract code and monitor public blockchain artifacts across multiple authorization sessions, including policy commitments, submitted proofs, public inputs, emitted events, calldata, timestamps, and gas usage. The adversary does not observe provider-to-TPS policy material, commitment-opening randomness, requester-submitted authorization evidence, credential witnesses, or TPS-side witness construction.

Each policy-enforcement instance corresponds to an authorization session, in which a requester seeks access to a protected resource under a committed policy. For each such session, the adversary observes a public view

$$view_{\mathsf{pub}} = (h, \pi, pp, meta), \tag{1}$$

where h is the on-chain commitment to the hidden policy, π is the zero-knowledge proof submitted for on-chain verification, pp denotes fixed-format public parameters or public handles, and $meta$ denotes transaction-level metadata exposed by the enforcement layer, such as calldata footprint, event structure, public-input layout, timing, and gas usage. The adversary's goal is to infer hidden information about the committed policy or the authorization interaction from this public view.

In addition to the public-chain observer, we treat the requester as untrusted with respect to policy confidentiality.

The requester may initiate access requests, observe its own submitted evidence and authorization outcome, and inspect the public transcript. However, the requester does not receive the encoded policy, commitment-opening randomness, or policy-bound proving material, which remain within the provider-to-TPS proof-generation boundary.

C. Privacy Objectives

We define policy privacy through three attack-oriented objectives.

Policy Content Hiding (PCH). PCH requires that public-chain artifacts should not reveal the semantic content of the hidden consent policy, such as data-use terms, disease categories, research purposes, institutional restrictions, or ethics-related requirements.

Policy Indistinguishability (PI). PI requires that different candidate policies should not become distinguishable from their public verification artifacts when they are compiled into the same fixed public interface. This captures whether two policies can be told apart from commitments, proofs, public parameters, or metadata even when their direct representations are hidden.

Structure and Parameter Privacy (SPP). SPP requires that public-chain artifacts should not reveal structural or parametric properties of the hidden policy, such as logical structure, nesting depth, predicate count, set size, or threshold-like parameters.

D. Problem Statement

Given a provider-defined consent policy P, requester authorization evidence, and an untrusted public blockchain, the problem is to enforce access to a protected healthcare resource while satisfying three requirements. First, authorization must be publicly verifiable: the ACM should be able to verify that the requester satisfies the committed policy without learning or executing the policy in plaintext. Second, policy-dependent information should remain hidden from unauthorized observers, including policy content, policy distinguishability, structural properties, requester evidence, and metadata-level enforcement fingerprints. Third, authorization outcomes should remain auditable, so that successful access decisions can be recorded and checked without exposing the hidden policy logic.

IV. TPS-ZKPI Design

A. Overview

TPS-ZKPI follows a Trusted Proof Service-assisted architecture for policy-private and auditable consent enforcement. As shown in Figure 1, the workflow binds a provider-committed hidden policy to requester-submitted authorization evidence through TPS-side proof generation and on-chain proof verification. The provider registers only a policy commitment on-chain and sends policy-bound proving material to the TPS off-chain. For each access request, the TPS constructs the private witness from the hidden policy material and requester evidence, generates a zero-knowledge proof of policy compliance, and submits a fixed-format proof statement for ACM verification. The ACM records the authorization outcome, which is then consumed by the resource access layer.

This design separates confidentiality-sensitive proof generation from public verification. Policy material and requester evidence remain within the provider–TPS–requester off-chain boundary, while the blockchain verifies only a fixed-format proof statement against the provider's policy commitment and session-binding public inputs.

B. Policy Representation and Interpretation

TPS-ZKPI represents heterogeneous Boolean-based consent policies in a fixed-length, circuit-friendly format. Let

$$X = \{0,1\}^N, \qquad Y = \{0,1\}^M \tag{2}$$

denote the requester-attribute space and the external-predicate space, respectively. The unified operand space is

$$O = X \times Y, \tag{3}$$

where $a \in X$ encodes requester attributes and $p \in Y$ encodes externally attested predicate outcomes, such as credential validity, ontology-linked checks, or obligation-related conditions. TPS-ZKPI does not itself solve credential issuance, revocation, or external attestation; it consumes these outcomes and, where applicable, verifies auxiliary witnesses inside the proof relation.

Consent policies are expressed as Boolean formulas:

$$P ::= atom \mid (P \wedge P) \mid (P \vee P) \mid \neg P. \tag{4}$$

Each policy is compiled into a Reverse Polish Notation (RPN) program over operand tokens, Boolean operators, and deterministic no-op tokens. To support reusable verification, each program is padded to a fixed length:

$$P_{enc} = [t_1, \ldots, t_{\mathsf{DEPTH}}], \tag{5}$$

where shorter policies are padded with no-op tokens. We denote the set of fixed-length programs by Prog.

The reusable policy interpreter is a zero-knowledge circuit

$$C : \mathsf{Prog} \times O \to \{0,1\}. \tag{6}$$

Given an encoded policy and operand values, the interpreter evaluates the RPN program using deterministic stack semantics and enforces well-formedness constraints inside the circuit. The circuit outputs 1 only if the program is syntactically valid and the requester evidence satisfies the encoded policy.

This design avoids policy-specific circuits and verifier contracts while supporting policy evolution through different fixed-length policy programs.

C. Commitment Binding and Proof Generation

To bind each proof to a provider-defined hidden policy, the provider samples commitment randomness r and computes

$$h = \mathsf{Com}(P_{enc}; r) = \mathsf{Poseidon}(P_{enc} \parallel r), \tag{7}$$

where h is stored on-chain as the public handle of the hidden policy. The commitment hides the policy representation while binding future proofs to the committed policy instance.

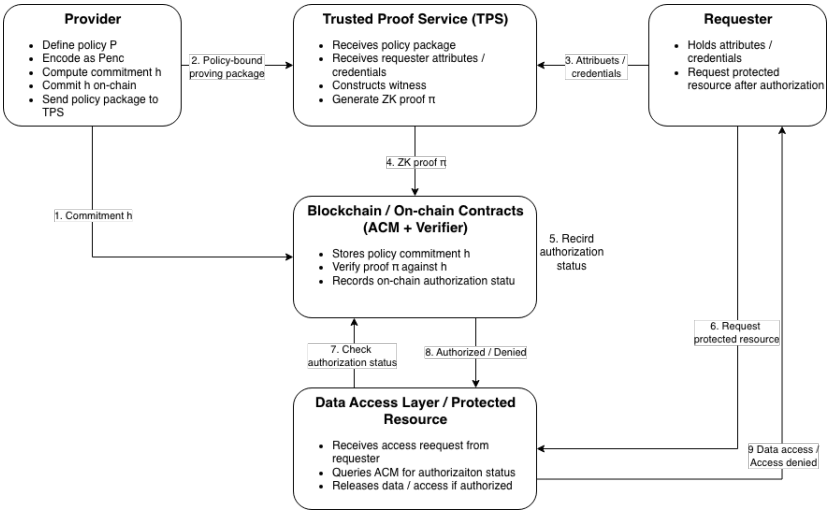


Fig. 1. TPS-assisted policy-compliance workflow.

For an access request, the TPS constructs the private witness

$$w = (P_{enc}, a, p, r, \omega), \tag{8}$$

where P_{enc} is the encoded policy, a denotes requester attributes, p denotes external predicate outcomes, r is the commitment randomness, and ω denotes auxiliary witnesses for credential- or predicate-dependent checks. The public statement contains the policy commitment h, requester and resource identifiers, a session nonce, and fixed-format public parameters pp.

The TPS generates a proof

$$\pi \leftarrow \mathsf{Prove}\big(C, w; (h, id_{req}, id_{res}, nonce, pp)\big). \tag{9}$$

The proof certifies that: (i) the encoded policy is well formed, (ii) the requester evidence satisfies the policy under the interpreter, (iii) the hidden policy encoding is consistent with the on-chain commitment h, (iv) required credential or external-predicate checks are satisfied with respect to the supplied parameters, and (v) the proof is bound to the requester, resource, and session context.

The ACM verifies π on-chain against the public statement. If verification succeeds, it records a positive authorization outcome for the requester and resource. The policy encoding, commitment randomness, requester attributes, credential witnesses, and auxiliary proof material are not posted on-chain.

D. Transcript Normalization

Zero-knowledge proofs hide private witness values, but surrounding blockchain artifacts may still leak policy-dependent information. TPS-ZKPI therefore normalizes the public transcript so that different policies induce fixed or tightly bounded observable profiles. Table II summarizes the mechanisms.

These mechanisms do not eliminate all public observability. Rather, they reduce policy-dependent variation in calldata, logs, public parameters, and gas-related behavior that could otherwise serve as a fingerprint for the metadata-only adversary defined in Section III.

TABLE II
TRANSCRIPT-NORMALIZATION MECHANISMS IN TPS-ZKPI.

Artifact	Normalization mechanism
Calldata	Fixed verification interface and padded proof/public-input encoding.
Logs	Fixed event structure with constant topic and payload layout.
Public parameters	Fixed-format handles instead of policy artifacts or credential witnesses.
Gas usage	Policy-independent verifier logic with bounded or padded gas profiles.

V. Privacy Evaluation

We evaluate the three privacy properties of our system—Policy Content Hiding, Policy Indistinguishability, and Structure & Parameter Privacy—against a passive on-chain observer (Section V-A). The primary evidence is a supervised, multi-seed attack evaluation with policy-disjoint splits and permutation tests (Section V-B); unsupervised clustering provides a label-free cross-check on the same executions. All three properties pass the supervised criterion, and the clustering attack finds no natural grouping by hidden labels.

A. Adversary and observables

We evaluate against a *passive on-chain observer*: an adversary who sees every transaction and event our protocol emits, but never the plaintext policy, the witness, or any client-side timing. This is the strongest realistic attacker for a public blockchain—it has unrestricted, perpetual read access to the ledger.

Concretely, for each policy execution (a commit transaction followed by an access transaction) the adversary observes the following nine on-chain features:

- **Gas:** `commitGasUsed`, `accessGasUsed`.
- **Calldata size:** `commitCalldataLength`, `accessCalldataLength`.
- **Proof size:** `proofSize`.
- **Event logs:** `commitLogsCount`, `commitLogsSize`, `accessLogsCount`, `accessLogsSize`.

- **Timing:** `blockDifference` (blocks between commit and access).

By design, calldata length, proof size, and log counts/sizes are content-independent (constant across policies), so any residual signal can only flow through the gas features. The defenses (commitment-only interface, randomized access-gas padding) target exactly this channel.

We consider three increasingly fine-grained attacker tasks, one per privacy property: (i) recover the policy's *content* (PCH, Section V-C); (ii) tell two *semantically related* policies apart (PI, Section V-D); and (iii) recover the policy's *structure or parameter magnitude* (SPP, Section V-E). The primary attacker is supervised and label-aware; a separate unsupervised clustering attacker provides a label-free cross-check (Section V-B).

B. Attack methodology

We evaluate each privacy property with a primary supervised attack and a complementary unsupervised cross-check. The *supervised* pipeline asks whether a trained classifier can recover a hidden label from the observables; this is the main privacy verdict because it gives the adversary labels and therefore more power than a purely exploratory analysis. The *unsupervised* pipeline asks whether the observables cluster along the hidden label without any supervision, providing a label-free sanity check on the same executions.

1) Supervised attacks: For every task we train linear and non-linear classifiers and report the score used by the unified verdict for that property. Each task is evaluated over **30 random seeds** using a **policy-disjoint train/test split**: the policies used to train the attacker never appear in the test set, so a high score cannot come from memorizing individual policies. For each seed we run a label-permutation test and record whether the AUC is significant at α=0.05.

The honest headline metrics are the AUC, its distance from chance $|\mathrm{AUC} - 0.5|$, and the permutation p-value. Raw accuracy advantage is retained only as a legacy diagnostic field because, on imbalanced test sets, a majority-class predictor can score large apparent gains with $F_1 \approx 0$. The unified verdict (`experiments/core/multiseed_criterion.js`) is:

> **FAIL** iff $\mathrm{AUC}_{\mathrm{mean}} > 0.55$ **and** the significant-seed fraction is ≥ 0.5 (over 30 policy-disjoint seeds); **PASS** otherwise.

Both legs are required: high classification power that is not statistically reproducible across seeds (or reproducible significance with chance-level AUC) does not constitute a leak. Subclasses whose median test size is $n < 20$ are *underpowered*—reported but excluded from gating, so a tiny subclass's noise cannot decide the verdict. The result tables report the AUC selected by the committed verdict file for each task. SPP uses the calibrated random-forest score for its AUC leg because logistic regression over-separates on the near-constant SPP features and yields a degenerate AUC.

2) Unsupervised clustering attacks: The clustering attacker drops the labels entirely and asks whether the gas features *naturally* group along the hidden property. We run K-Means (with K set to the number of classes, or K=2 for policy pairs) and DBSCAN on the on-chain gas features, and score the recovered clusters against the true labels with the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI). Values near 0 mean the clusters carry no information about the hidden label; values near 1 mean full leakage.

TABLE III
HEADLINE PRIVACY RESULTS ON THE POLYGON AMOY DATASET. THE SUPERVISED ATTACKER USES 30 POLICY-DISJOINT SEEDS AND A PERMUTATION TEST. A TASK PASSES UNLESS BOTH $\mathrm{AUC}_{\mathrm{MEAN}} > 0.55$ AND AT LEAST 50% OF SEEDS ARE SIGNIFICANT.

Property	Hidden target	Headline supervised result	Verdict
PCH	DUO terms and ICD-10 chapters	16 content-recovery tasks pass; DUO AUCs 0.490–0.527, ICD AUCs 0.458–0.573	✓
PI	Related policy-pair identity	Overall AUC 0.516 ± 0.030, permutation $p = 0.367$, 10% significant seeds	✓
SPP	Structure, predicate, and parameter buckets	18 pairwise contrasts pass; AUCs 0.428–0.521, max 27% significant seeds	✓

PCH clusters all executions globally (no per-layer splitting), so the large-sample ARI/NMI are read directly. For the PI pairs we additionally guard against two small-sample failure modes.

a) Controlling sample-size confounds.: The naive PI pair construction produced severely imbalanced pairs (one policy with 1 execution against another with up to 30), where K-Means cannot cut the data and ARI/NMI ≈ 0 is an artifact rather than evidence of privacy. We therefore rebuild the PI pairs from policies with ≥ 30 executions into balanced 30-vs-30 contrasts before clustering. To rule out residual finite-sample inflation, we add a *permutation baseline*: we shuffle the pair labels 200 times and recompute the clustering NMI, giving the NMI distribution under zero true signal. The debiased NMI is the observed value minus this baseline, and the empirical p-value is the fraction of permutations whose NMI meets or exceeds the observed; a pair leaks only if its debiased NMI is positive *and* significant ($p < 0.05$).

b) Dataset.: Both attack families run on the same fixed-circuit, commitment-only, access-gas-padded *Polygon Amoy* dataset: the supervised verdicts read `experiments/multiseed_results/polygonAmoy/` (numbers of record), and the clustering reads the same executions exported to `python/data/polygonAmoy/`. The supervised and unsupervised attacks therefore constitute two independent attack families applied to one dataset, rather than two datasets. Per-task sample sizes vary and are listed in the result tables (e.g. 2005 DUO executions; 133 balanced 30-vs-30 PI pairs); the PI debiasing averages each observed NMI over 20 K-Means seeds against a 200-permutation baseline. Provenance for every table and figure is listed in `paper/README.md`.

C. Policy Content Hiding (PCH)

a) Property.: PCH requires that the on-chain observer cannot recover the *content* of a policy—its DUO data-use

TABLE IV
POLICY CONTENT HIDING: PER-TERM SUPERVISED ATTACK OVER 30 POLICY-DISJOINT SEEDS. EACH DUO TERM IS A ONE-VS-REST BINARY TASK. *%sig* IS THE FRACTION OF SEEDS WHOSE PERMUTATION TEST REJECTS AT α=0.05. ALL SIX TASKS PASS THE UNIFIED CRITERION.

Term	AUC (mean $\pm$ std)	perm p	%sig	med n	verdict
DUO GR	0.497 ± 0.024	0.535	0%	613	✓
DUO DS	0.490 ± 0.021	0.624	3%	613	✓
DUO RS	0.527 ± 0.019	0.191	13%	613	✓
DUO POA	0.493 ± 0.023	0.579	0%	613	✓
DUO GS	0.521 ± 0.018	0.251	13%	613	✓
DUO CC	0.501 ± 0.018	0.479	0%	613	✓
Overall: ✓ PASS					

terms or the ICD-10 disease codes it references—better than chance.

b) Scope of the labels.: We do not attack the full Data Use Ontology and its linked ontologies. The complete DUO has 23 data-use terms, and ICD-10 has tens of thousands of individual codes; most never appear in our generated policies, and those that do are too rare to support a statistically powered one-vs-rest attack. We therefore evaluate the six DUO terms that occur with enough support in the dataset (GR, DS, RS, POA, GS, CC; smallest class n=30) and aggregate disease codes to the ten ICD-10 chapters. This is the right granularity for the privacy question—the observer cares whether the *category* of permitted use or disease leaks, not which leaf code is used—and it is also a conservative choice: attacks on coarser, better-supported labels are *easier* than on rare leaf terms, so passing here does not understate the leak. The defenses (commitment-only interface, gas padding) are content-agnostic and apply identically to the terms we do not enumerate.

c) Supervised attack.: We treat each of the six DUO terms (GR, DS, RS, POA, GS, CC) as a one-vs-rest binary recovery task and run the multi-seed pipeline of Section V-B. Table IV reports the authoritative per-term verdicts: every DUO task has $\mathrm{AUC}_{\mathrm{mean}} \in [0.490, 0.527]$ and a significant-seed fraction of at most 13%, so none approaches either leg of the failure criterion. The same pipeline applied to the ten ICD-10 chapters (one-vs-rest over the ICD-bearing rows only; Table V) passes all ten. The single highest value is ch6 at $\mathrm{AUC}_{\mathrm{mean}} = 0.573 \pm 0.061$ with 17% significant seeds; it clears the AUC leg ($0.573 > 0.55$) but not the reproducibility leg ($17\% < 50\%$), so it is a boundary case rather than a stable channel. We report such cases rather than suppress them.

D. Policy Indistinguishability (PI)

a) Property.: PI strengthens PCH: even when two policies are *semantically related*, the observer must not tell which one was executed—formally, $\Pr[\hat{b} = b] \approx \frac{1}{2}$. We test four pair archetypes: `semantic_equivalent` (identical normalized logic), `structural_variant` (same semantics, different execution protections), `weak_equivalent` (≥ 0.95 semantic overlap), and `controlled_difference` (small targeted deltas; a regression control).

TABLE V
POLICY CONTENT HIDING: PER-CHAPTER ICD-10 SUPERVISED ATTACK OVER 30 POLICY-DISJOINT SEEDS (ONE-VS-REST, ICD-BEARING ROWS ONLY; MEDIAN TEST $n = 300$). THE HIGHEST VALUE IS CH6 (AUC 0.573, %SIG 17%), BUT IT CLEARS NEITHER THE AUC NOR THE REPRODUCIBILITY LEG OF THE CRITERION, SO ALL TEN CHAPTERS PASS.

Chapter	AUC (mean $\pm$ std)	perm p	%sig	med n	verdict
ICD ch0	0.488 ± 0.068	0.536	0%	300	✓
ICD ch1	0.490 ± 0.050	0.560	0%	300	✓
ICD ch2	0.513 ± 0.051	0.424	7%	300	✓
ICD ch3	0.524 ± 0.042	0.358	0%	300	✓
ICD ch4	0.547 ± 0.055	0.278	17%	300	✓
ICD ch5	0.458 ± 0.060	0.700	0%	300	✓
ICD ch6	0.573 ± 0.061	0.206	17%	300	✓
ICD ch7	0.490 ± 0.048	0.595	0%	300	✓
ICD ch8	0.537 ± 0.055	0.327	10%	300	✓
ICD ch9	0.524 ± 0.057	0.395	7%	300	✓
Overall: ✓ PASS (all 10 chapters)					

b) Supervised attack.: The distinguisher's task is to decide, from the on-chain features of a single execution, which policy of the pair produced it. We evaluate this single-sample classifier over 30 policy-disjoint seeds (Table VI). The overall AUC is 0.516 ± 0.030 with permutation $p = 0.367$ and only 10% of seeds significant. All three gating archetypes—`semantic_equivalent`, `structural_variant`, and `weak_equivalent`—remain within noise of chance. `controlled_difference` is reported as a diagnostic regression control but does not gate the PI verdict.

c) Why multi-seed matters.: Earlier three-seed diagnostics made the `structural_variant` archetype look separable, but the 30-seed policy-disjoint run in Table VI gives AUC 0.524 ± 0.071 with 0% significant seeds. This is precisely the failure mode the multi-seed criterion of Section V-B is designed to reject: a single split can create an apparently meaningful spike that does not reproduce across independent policy splits.

E. Structure & Parameter Privacy (SPP)

a) Property.: SPP requires that neither the *shape* of a policy (logical structure, nesting depth, predicate coverage) nor the *magnitude* of its parameters (whitelist/set sizes) leaks to the observer. The supervised attack organizes the test along three annotation axes—logical structure type, predicate count, and parameter magnitude—and attacks each pair of buckets within an axis.

b) Supervised attack.: Table VII reports all 18 pairwise contrasts over 30 policy-disjoint seeds across `structureType`, `predicateCombo`, and `paramType`. AUC is the calibrated random-forest score, since logistic regression over-separates on the near-constant SPP features. Every contrast has $\mathrm{AUC}_{\mathrm{mean}} \in [0.428, 0.521]$ and a significant-seed fraction at or below 27%, so all 18 pass and SPP passes overall. We gate on AUC and the permutation test. Single-split accuracy-style diagnostics can run large on imbalanced contrasts, but this is the majority-class artifact described in Section V-B—a near-constant predictor scoring apparent gains without genuine separation—which the AUC-plus-permutation criterion correctly reads as chance.

TABLE VI
POLICY INDISTINGUISHABILITY: SINGLE-SAMPLE DISTINGUISHER (LEVEL I) OVER 30 POLICY-DISJOINT SEEDS, BY POLICY-PAIR ARCHETYPE. CONTROLLED_DIFFERENCE IS REPORTED AS A DIAGNOSTIC REGRESSION CONTROL AND EXCLUDED FROM THE PI VERDICT.

Pair archetype	AUC (mean ± std)	adv (mean ± std)	perm p	%sig	med n	verdict
overall	0.516 ± 0.030	0.026 ± 0.022	0.367	10%	270	✓
semantic_equivalent	0.511 ± 0.082	0.070 ± 0.045	0.456	13%	71	✓
structural_variant	0.524 ± 0.071	0.059 ± 0.046	0.407	0%	40	✓
weak_equivalent	0.494 ± 0.085	0.069 ± 0.051	0.529	7%	54.5	✓
controlled_difference	0.510 ± 0.076	0.065 ± 0.041	0.452	3%	68	– (diagnostic)
Overall: ✓ PASS						

TABLE VII
STRUCTURE & PARAMETER PRIVACY: 18 PAIRWISE CONTRASTS ACROSS THREE ANNOTATION AXES, EACH OVER 30 POLICY-DISJOINT SEEDS. AUC IS THE CALIBRATED RANDOM-FOREST SCORE. EVERY CONTRAST SATISFIES $\text{AUC}_{\text{MEAN}} \leq 0.55$ *and* %SIG $< 50\%$.

Contrast	AUC (mean ± std)	adv (mean ± std)	perm p	%sig	med n
structureType					
and_only / or_only	0.475 ± 0.051	0.036 ± 0.029	0.547	10%	102
and_only / negation_heavy	0.428 ± 0.058	0.060 ± 0.028	0.537	23%	90
and_only / and_or_mixed	0.521 ± 0.073	0.057 ± 0.041	0.432	27%	78
or_only / negation_heavy	0.440 ± 0.061	0.049 ± 0.030	0.491	20%	81
or_only / and_or_mixed	0.504 ± 0.066	0.044 ± 0.037	0.520	17%	66
negation_heavy / and_or_mixed	0.515 ± 0.076	0.049 ± 0.031	0.548	13%	57
predicateCombo					
none / single	0.467 ± 0.073	0.041 ± 0.033	0.571	0%	54
none / few	0.484 ± 0.061	0.032 ± 0.022	0.525	10%	102
none / many	0.511 ± 0.080	0.067 ± 0.036	0.595	7%	42
single / few	0.466 ± 0.053	0.039 ± 0.030	0.645	0%	117
single / many	0.476 ± 0.064	0.049 ± 0.036	0.537	13%	54
few / many	0.521 ± 0.059	0.043 ± 0.034	0.535	10%	102
paramType					
none / small	0.480 ± 0.074	0.050 ± 0.041	0.628	3%	54
none / medium	0.458 ± 0.074	0.056 ± 0.037	0.540	13%	96
none / large	0.521 ± 0.062	0.055 ± 0.037	0.569	3%	48
small / medium	0.513 ± 0.052	0.030 ± 0.025	0.583	3%	108
small / large	0.486 ± 0.072	0.046 ± 0.031	0.573	10%	63
medium / large	0.493 ± 0.068	0.036 ± 0.028	0.531	20%	102
Overall: ✓ PASS (all 18 contrasts)					

F. Unsupervised clustering: a label-free cross-check

The supervised attacks (Sections V-C–V-E) assume an adversary who can train on labeled executions. A weaker but assumption-free attacker simply asks whether the on-chain gas features *cluster* along the hidden property with no supervision at all. Table VIII consolidates this attack across all three properties: K-Means recovers essentially nothing, with ARI and NMI within ± 0.02 of zero for every DUO, PI, and SPP target. The unsupervised attacker agrees with the supervised verdict—privacy holds—as a second, label-free attack family on the same dataset (Figures 2 and 3).

a) Removing sample-size confounds.: The clustering inputs are constructed to avoid degenerate measurements. The naive PI pairs were severely imbalanced (executions in a 1:30 ratio), under which K-Means trivially fails and a low ARI/NMI would be meaningless. We rebuild the pairs into balanced 30-vs-30 contrasts, and guard the result with a permutation baseline—NMI under shuffled labels, where no true signal can exist—reporting the debiased NMI (observed minus baseline) and its empirical p-value. The conclusion is unchanged (Section V-D): the indistinguishability is a property of the protocol, not of a degenerate measurement.

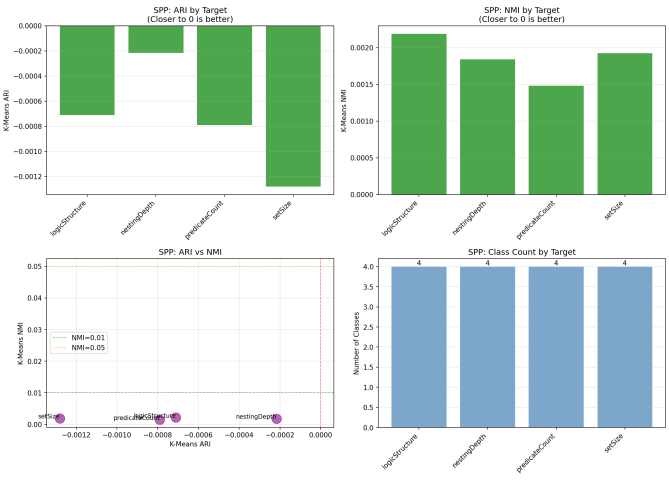


Fig. 2. SPP unsupervised clustering: ARI and NMI by structural target. All four targets sit at the zero line—gas clusters carry no structure or parameter information.

G. Sample-size sensitivity of NMI

The clustering cross-check (Section V-F) treats ARI as the primary conservative indicator because NMI can take moder-

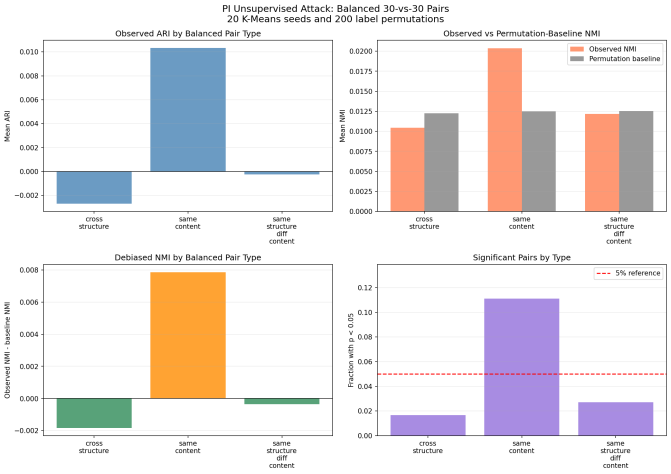


Fig. 3. PI unsupervised clustering by pair archetype: per-pair ARI/NMI stay at chance across all archetypes, matching the supervised distinguisher.

TABLE VIII
UNSUPERVISED CLUSTERING ATTACK (K-MEANS, K=NUMBER OF CLASSES / K=2 FOR PAIRS) ON THE ON-CHAIN GAS FEATURES. ARI AND NMI NEAR 0 MEAN THE NATURAL GAS CLUSTERS CARRY NO INFORMATION ABOUT THE HIDDEN LABEL. PCH CLUSTERS ALL EXECUTIONS GLOBALLY (NO LAYERING).

Property	Target	n	#cls.	ARI	NMI
PCH	DUO terms	2005	6	0.0002	0.0036
PCH	ICD-10 chapters	920	10	−0.0010	0.0181
PI	policy pairs (30v30)	133	2	0.0015	0.0136
SPP	logicStructure	1141	4	−0.0007	0.0022
SPP	nestingDepth	1141	4	−0.0002	0.0018
SPP	predicateCount	1141	4	−0.0008	0.0015
SPP	setSize	1141	4	−0.0013	0.0019

ately high values under small samples. To justify that choice, we quantify how *finite-sample effects* alone can inflate NMI even when there is *no* discriminative signal, using an idealized null in which all classes are statistically indistinguishable.

a) Null hypothesis.: We sample every class from the *same* distribution, $x_i \sim \mathcal{N}(0, I_d)$ for all classes. Since no class carries class-specific information, a clustering algorithm should not recover class membership, and both NMI and ARI should concentrate around 0 as the sample size grows.

b) Simulation.: For each configuration we run $T = 1000$ independent trials. In each trial we (i) generate C classes with n i.i.d. samples per class from $\mathcal{N}(0, 1)$, (ii) apply K-Means with $K = C$ to the pooled data, and (iii) score the predicted assignments against the ground-truth labels with NMI and ARI. We sweep n while holding C fixed, and report the trial means $\overline{\text{NMI}}$ and $\overline{\text{ARI}}$. We consider a **PI-like** binary structure ($C = 2$, $n \in \{3, 10, 50, 100\}$) and an **ICD10-like** multi-class structure ($C = 5$, $n \in \{3, 10, 100\}$).

c) Results.: Table IX shows that, although all classes are indistinguishable by construction, **NMI is noticeably positive at very small n but collapses toward zero as n grows**, whereas **ARI stays near zero throughout**. For the binary structure, $\overline{\text{NMI}}$ falls from 0.1808 at n=3 (N=6) to 0.0404 at n=10, and to 0.0066/0.0038 at n=50/100, while $\overline{\text{ARI}}$ stays at −0.0190, −0.0020, −0.0011, 0.0002. For the multi-class structure the same trend holds: $\overline{\text{NMI}}$ goes from 0.4168 (n=3) to 0.1249 (n=10) to 0.0105 (n=100), with $\overline{\text{ARI}}$ near zero (0.0005, 0.0018, −0.0000). Hence moderately high NMI in small-sample clustering should not be read as actionable leakage, and ARI is the more conservative indicator of practical recoverability in our unsupervised evaluation.

TABLE IX
SAMPLE-SIZE SENSITIVITY OF NMI AND ARI UNDER THE NULL HYPOTHESIS WHERE ALL CLASSES ARE I.I.D. SAMPLED FROM $\mathcal{N}(0, 1)$ (T=1000 TRIALS PER ROW).

Structure	n (per class)	C	N	$\overline{\text{NMI}}$	$\overline{\text{ARI}}$
PI-like	3	2	6	0.1808	−0.0190
PI-like	10	2	20	0.0404	−0.0020
PI-like	50	2	100	0.0066	−0.0011
PI-like	100	2	200	0.0038	0.0002
ICD10-like	3	5	15	0.4168	0.0005
ICD10-like	10	5	50	0.1249	0.0018
ICD10-like	100	5	500	0.0105	−0.0000

H. Ablation: contribution of access-gas padding

The main results show that the full mechanism hides content, policy identity, and structure/parameters. The ablation asks *why*: how much of the residual attack surface is removed by randomized access-gas padding after the fixed circuit and commitment-only interface have already eliminated calldata, proof-size, and event-log channels.

We compare two aligned protection profiles on the same selected policy set. **B1** uses the fixed circuit, commitment-only interface, and TPS execution profile without commit/access gas padding. **B2** is the full mechanism used in the main evaluation, with randomized access-gas padding. This paired design also validates the attack pipeline directly: the same features, splits, and classifiers produce well-above-chance AUCs on B1 when padding is absent, while B2 returns the attacker toward chance. Thus the main privacy claim rests on a real ablation contrast rather than on a synthetic label derived from an input feature. The paired leakage metric is

$$\Delta_{\text{leak}} = |\text{AUC}_{\text{B1}} - 0.5| - |\text{AUC}_{\text{B2}} - 0.5|.$$

Positive values mean padding moved the attacker closer to chance, and we bound each Δ_{leak} with a 500-iteration bootstrap 95% confidence interval. Table X is intentionally separated from the main privacy verdicts: the main verdicts answer whether the full mechanism is private; this ablation attributes that privacy to a specific defense layer and reports its gas cost.

a) Padding removes otherwise-recoverable signal.: With padding disabled, attacks that are non-informative on the deployed system become measurably effective. The random-forest distinguisher on B1 reaches AUC up to 0.81 on SPP parameter-magnitude contrasts (`paramType none/small`) and 0.76 on structural contrasts (`and_only/negation_heavy`), while the corresponding B2 AUCs fall back to 0.41–0.47; the bootstrap Δ_{leak} is positive with a 95% CI that excludes zero. The clustering attacker shows the same pattern: PI pairwise clustering reaches debiased significance on 68% of pairs

TABLE X
GAS-PADDING ABLATION ON POLYGON AMOY. B1 (NO COMMIT/ACCESS GAS PADDING) AND B2 (DEPLOYED, RANDOMIZED ACCESS-GAS PADDING) ARE ALIGNED ON THE SAME POLICY SET. SUPERVISED AUC IS THE CALIBRATED RANDOM-FOREST SCORE; CLUSTERING REPORTS K-MEANS ARI. $\Delta_{\text{LEAK}} = |\text{AUC}_{\text{B1}} - 0.5| - |\text{AUC}_{\text{B2}} - 0.5| > 0$ MEANS PADDING MOVED THE ATTACKER TOWARD CHANCE; Δ_{LEAK} ROWS SHOW THE BOOTSTRAP MEAN WITH 95% CI (500 ITERATIONS). PADDING COLLAPSES EVERY RECOVERABLE SIGNAL TO THE CHANCE BASELINE.

Attack	Target	B1 (unpad.)	B2 (pad.)
Supervised (RF AUC; Δ_{leak} with 95% CI)			
SPP	`paramType none/small` $\Delta_{\text{leak}} = 0.127$ CI$[0.055, 0.201]$	0.812	0.414
SPP	`structure and/neg-heavy` $\Delta_{\text{leak}} = 0.195$ CI$[0.173, 0.217]$	0.759	0.465
PCH	DUO POA $\Delta_{\text{leak}} = 0.092$ CI$[0.071, 0.114]$	0.616	0.491
PI	`controlled_difference` $\Delta_{\text{leak}} = 0.102$ CI$[0.076, 0.129]$	0.611	0.515
Unsupervised (K-Means ARI)			
PCH	ICD-10 chapter	0.064	−0.005
SPP	logic structure	0.067	−0.006
SPP	predicate count	0.054	0.001
Gas cost (mean, B1 → B2)			
Commit gas		216,562	303,556 (+40.2%)
Access gas		546,876	753,310 (+37.7%)

under B1 (mean ARI 0.50) versus 5% under B2 (mean ARI 0.0015), and SPP logic-structure clustering drops from ARI 0.067 (B1) to −0.006 (B2). This establishes randomized access-gas padding as the defense layer responsible for the protection reported in Sections V-C–V-F, rather than weak attacks or insufficient sample size.

b) Cost.: Padding is not free: it raises mean commit gas from 216,562 (B1) to 303,556 (B2), a 40.2% increase, and mean access gas from 546,876 to 753,310, a 37.7% increase. This overhead is the principal efficiency tradeoff of the design.

c) Scope.: B2 reuses the existing deployed Polygon Amoy run, so B1 and B2 are non-contemporaneous; `blockDifference` is excluded from every attack model to avoid confounding the comparison with timing rather than policy-dependent structure.

VI. DISCUSSION

The evaluation indicates that TPS-ZKPI removes recoverable policy-dependent signals from normalized public-chain artifacts. Across all PCH, PI, and SPP tasks, supervised classifiers stay below the multi-seed permutation-test leakage criterion, and clustering ARI is statistically indistinguishable from zero. The gas-padding ablation (Section V-H) sharpens this conclusion: the same attacks become measurably effective once normalization is disabled, which attributes the protection specifically to transcript normalization rather than to weak attacks or insufficient sample size. Together with fixed-length policy representation, reusable interpretation, and commitment binding, these mechanisms limit the extent to which public-chain observers infer hidden policy content or structural properties from authorization transactions.

These results support a scoped privacy claim. The evaluation focuses on a metadata-only public-chain adversary that observes commitments, proofs, public parameters, calldata, events, and gas-related artifacts, but not TPS-side witness construction or off-chain policy material. Within this scope, transcript normalization eliminates the recoverable policy-dependent signals we are able to measure across PCH, PI, and SPP, at a gas-cost overhead of roughly 38–40%. We do not claim distinguishability is information-theoretically zero; rather, no modeled attacker recovers it above the chance baseline. Stronger adversaries, such as compromised proof services, adaptive requester-side probing, or off-chain side channels, remain outside the current evaluation scope and motivate the reduced-trust extensions discussed below.

The TPS trust assumption is the main architectural tradeoff. By moving confidentiality-sensitive proof generation to the TPS, TPS-ZKPI avoids exposing policy-bound proving material to requesters or public smart contracts, while preserving on-chain verifiability through zero-knowledge proof verification. However, this shifts part of the confidentiality burden to the service layer. Future work should investigate reduced-trust TPS designs, including distributed proof services, hardware-assisted isolation, accountable logging, or cryptographic protocols that further reduce what any single proof service can observe.

VII. CONCLUSION

This paper studied policy-private consent enforcement in blockchain-based healthcare data sharing under public-chain observability. We proposed TPS-ZKPI, a Trusted Proof Service-assisted zero-knowledge approach that binds hidden provider-defined policies to on-chain commitments, evaluates Boolean-based consent policies through a reusable policy interpreter, and normalizes public verification artifacts to reduce metadata-level leakage. Across supervised and unsupervised attacks on PCH, PI, and SPP, no modeled adversary recovered hidden policy information above the chance baseline under a multi-seed permutation criterion, and a paired gas-padding ablation attributed this protection specifically to transcript normalization—at a 38–40% gas overhead. Future work will extend the analysis to stronger adaptive adversaries, off-chain leakage, richer policy languages, reduced-trust proof generation, and privacy-preserving resource discovery.

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